this need not always be the case. Frequently, the abbreviation may be formed from the original word by skipping a few letters, as in the case of 'msg' which stands for 'message' or 'pt' which can stand for 'patient'. We run into the abbreviation sense disambiguation problem in the case of such non-standard abbreviations also. Consider the abbreviation 'mg'. It can stand for either magnesium or for milligram.

As the second step in the abbreviation expansion process, all the potential expansions are identified. In the third step, different ranking heuristics are applied to find the best match for the abbreviation under consideration. Depending on the context, abbreviation sense disambiguation can hopefully determine the correct expansion. Different heuristics are employed to compute the potential candidate set for each abbreviation. As mentioned earlier, since most documents typically contain the abbreviation co-located with the expansion at the point of definition, it is possible to build an approximate pattern matcher which scans the document for such occurrences, and builds up a dictionary of abbreviation and expansion phrases.

Given that many abbreviations are widely used and are part of 'external world knowledge', it is possible to use external knowledge bases to build a dictionary of abbreviations and expansions. For instance, one can use Wikipedia or other such knowledge bases to build up an abbreviation dictionary. Such dictionaries are especially useful for domain-specific abbreviation expansion. Once the list of candidate expansion phrases is found, post-processing or ranking can be employed to choose the best matched expansion for each abbreviation. This can be performed by using the context of the text in which the abbreviation appears, using supervised machine learning approaches.

Now let us look at the problem of abbreviation expansion in clinical texts. The biggest challenge with clinical abbreviations is that they are typically non-standard. Also, the expansion phrase usually does not co-occur with the clinical text. Certain abbreviations are technical and standard in nature, such as CABG which stands for 'Coronary Artery Bypass Graft' and can be easily determined by using an abbreviation dictionary. But many other abbreviations are non-standard. For instance, one frequently sees the abbreviation 'bs' for blood sugar. Also, even within the same clinical text, it is possible to associate multiple senses with the same abbreviation. For instance, consider the abbreviation 'ARF'. It could potentially stand for Acute Respiratory Failure or Acute Renal Failure. The nature of the document, i.e., it being a clinical progress note, does not help us in any way to distinguish between the two different senses for this abbreviation.

There has been considerable work in analysing clinical texts for abbreviation expansion. Many of them use a dictionary based approach, where the dictionary is built for a given corpus either manually or automatically, using a supervised machine learning approach. Expanding of non-standard abbreviations such as 'msg' can be modelled as an edit distance problem. Edit distance algorithms have been widely used for spelling correction. Here, we consider the source abbreviation and each target expansion, and compute the edit distance to morph the target expansion to the source abbreviation.

Here is a question for our readers. In general edit distance problems, we typically choose the least cost edit distance target word present in the dictionary as the correct term that was intended by the user. Would this choice be appropriate for abbreviation expansion? If not, what would be an appropriate edit distance heuristic for abbreviation expansion? We will discuss this problem further in our next column.

If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at *sandyasm_AT_yahoo_DOT_com*. Till we meet again next month, happy programming! END

By: Sandya Mannarswamy

The author is an expert in systems software and is currently in that happy space one enters when between jobs. Her interests include compilers, storage systems and natural language processing. If you are preparing for systems software interviews, you may find it useful to visit Sandya's LinkedIn group *Computer Science Interview Training India* at *http://www.linkedin.com/groups?home=HYPERLINK "http://www.linkedin.com/groups?home=&gid=2339182"&HYPERLINK "http://www.linkedin.com/groups?home=&gid=2339182"gid=2339182*